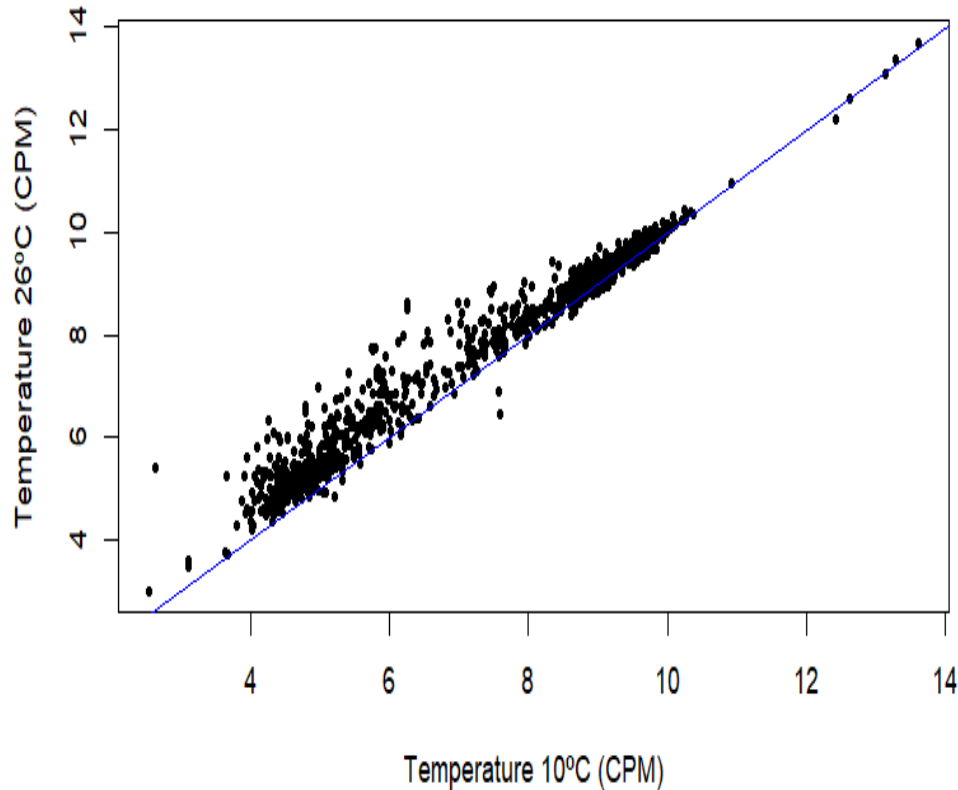


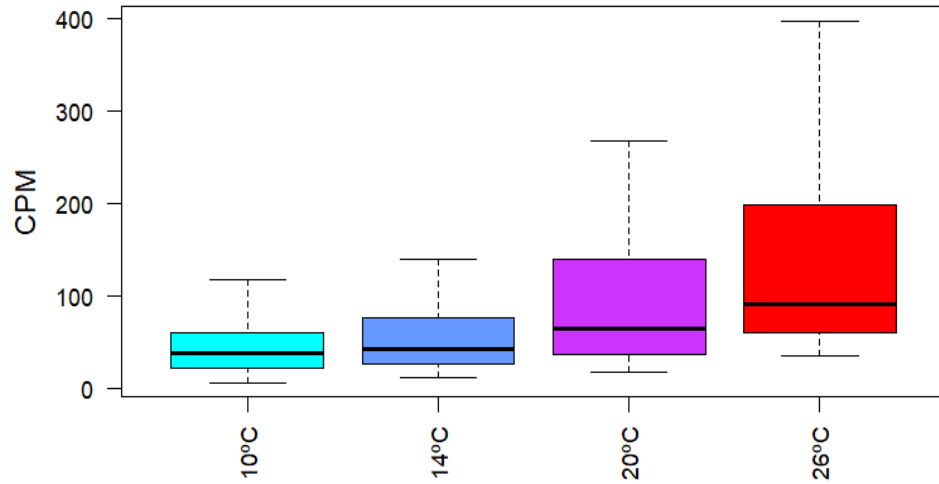
H3K4me3

26°C vs 10°C

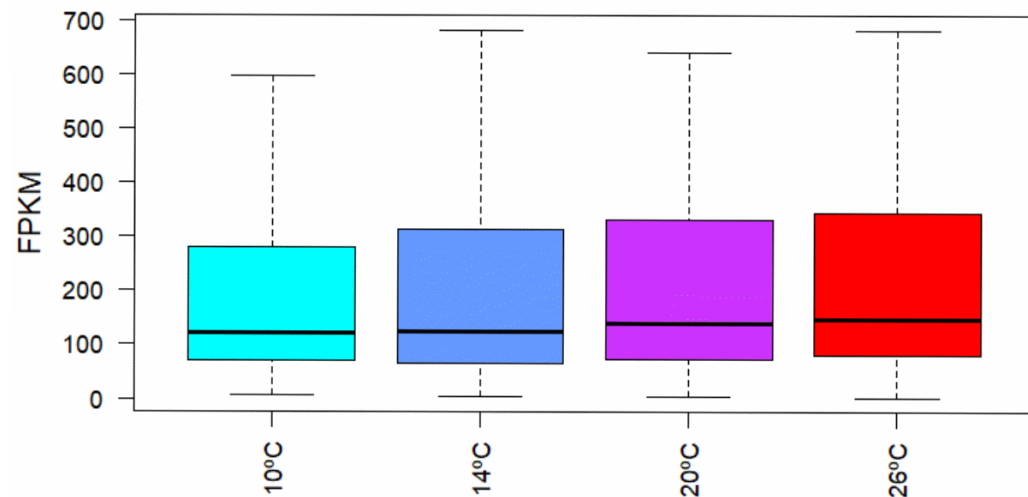
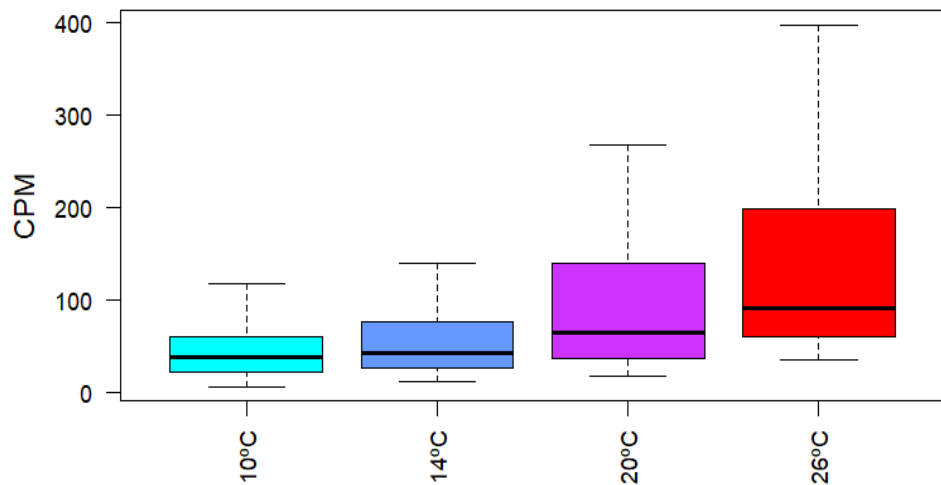


**83 peaks (7.85%) significantly
gain H3K4me3 as temperature
rises from 10°C to 26°C
according to a fold-change of 2
and a q-value of 0.05**

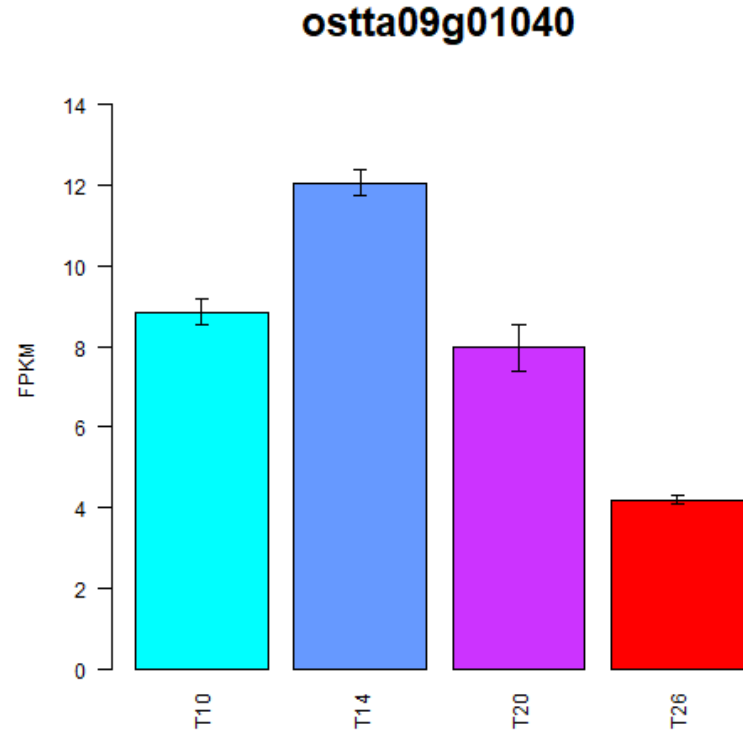
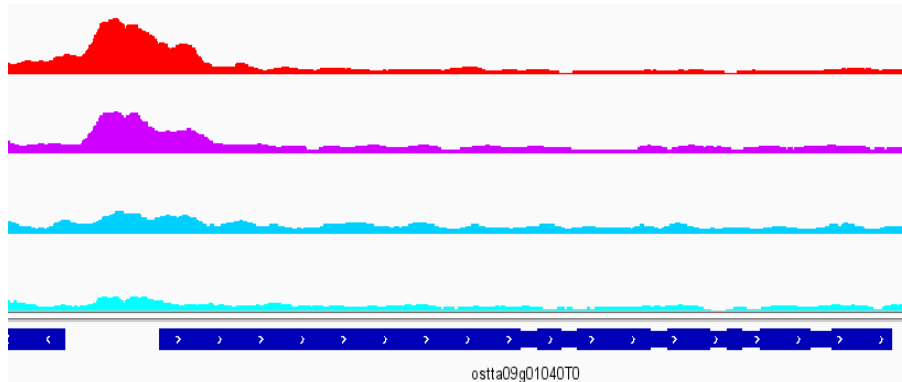
H3K4me3



H3K4me3



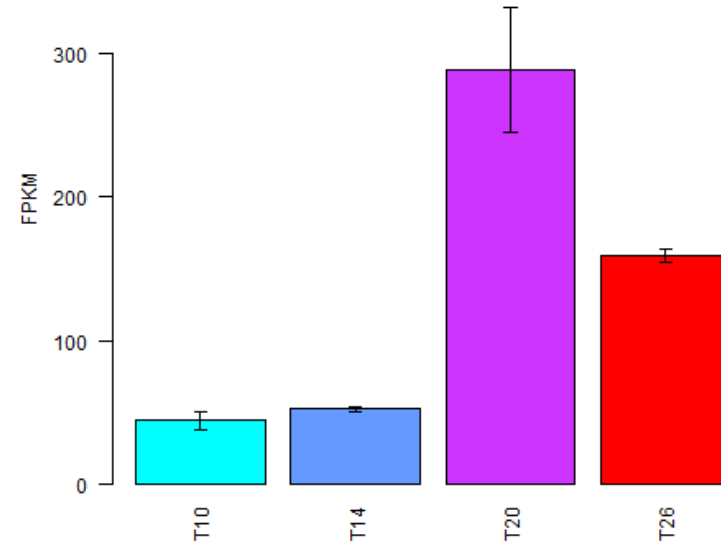
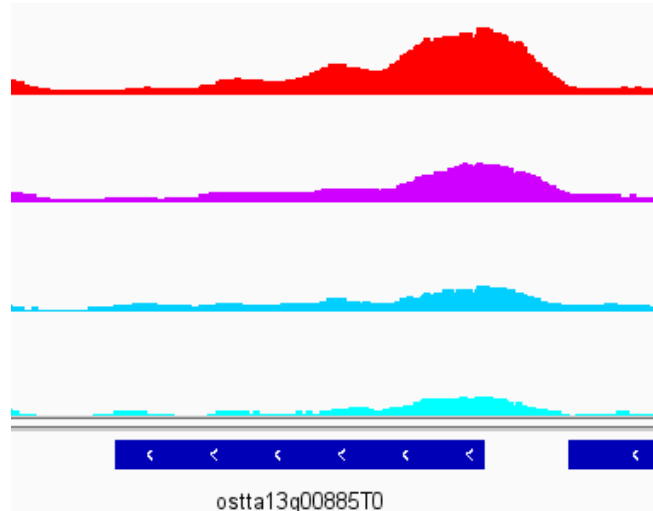
Protein Quality Control



- *ostta09g01040*: **Tetratricopeptide repeat**
- AT1G62740.1 Encodes one of the 36 carboxylate clamp (CC)-tetratricopeptide repeat (TPR) proteins (Prasad 2010, Pubmed ID: 20856808) with **potential to interact with Hsp90/Hsp70 as co-chaperones**. The mRNA is cell-to-cell mobile.

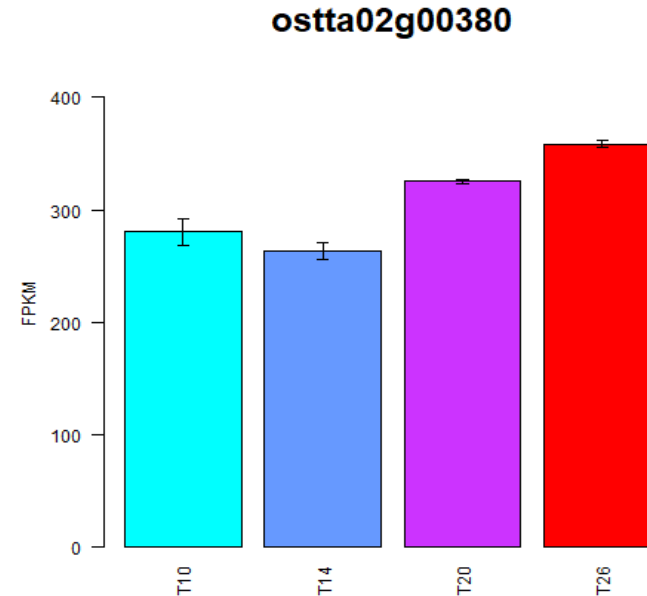
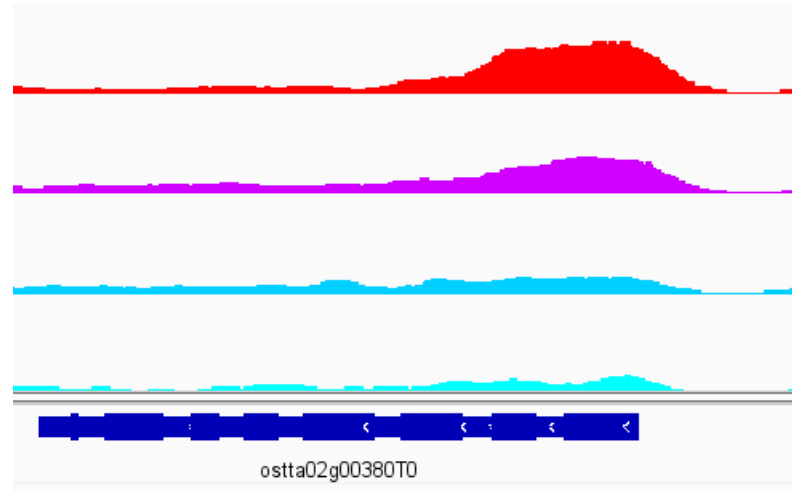
Protein Quality Control

osta13g00885



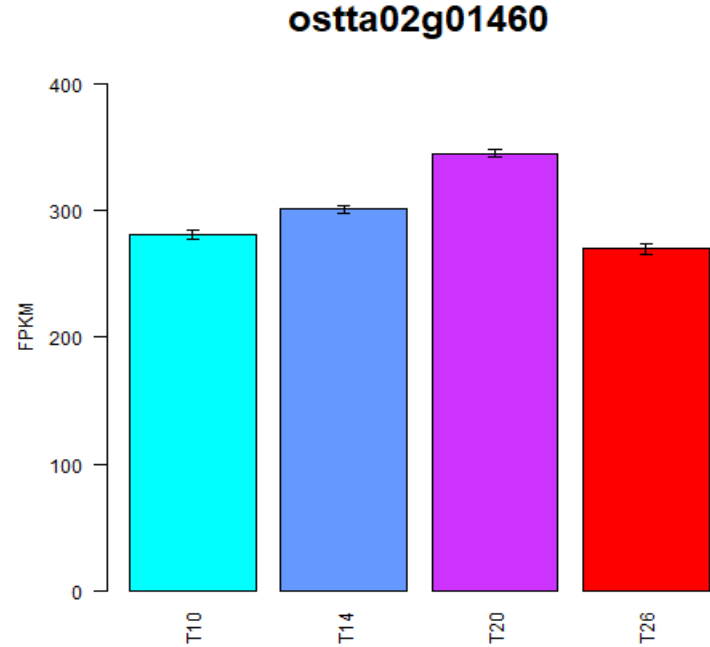
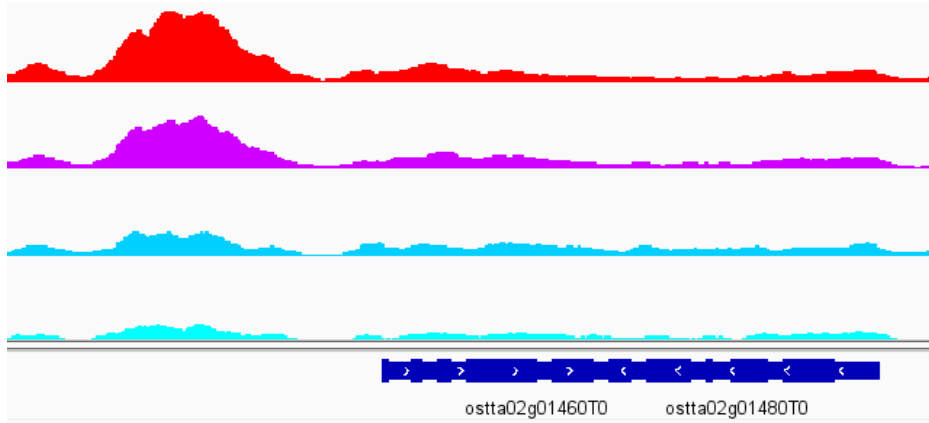
- *osta13g00885* hypothetical protein
- AT4G04330.1 Encodes a chloroplast thylakoid localized RbcX protein that **acts as a chaperone in the folding of Rubisco**.

Protein Quality Control



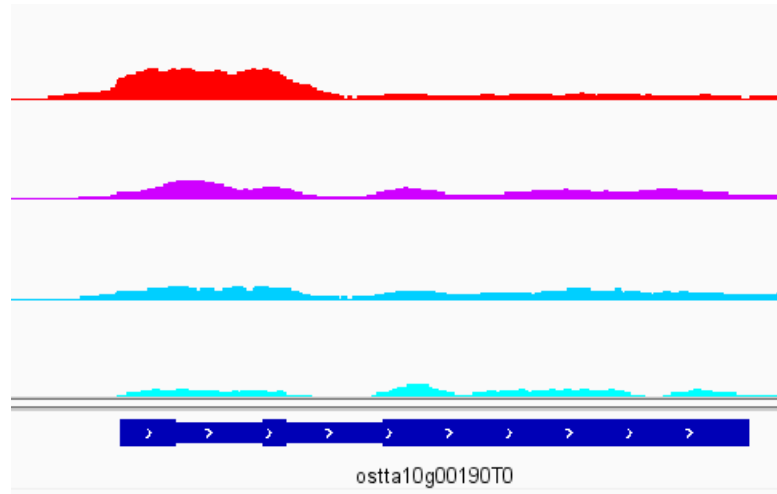
- *ostta02g00380* Clp protease proteolytic subunit /Translocation-enhancing protein TepA
- AT4G17040.1HON5 (At4g17040) encodes the ClpR4 subunit of the **chloroplast-localized Clp protease complex**. hon mutations disturb plastid protein homeostasis, thereby activating plastid signaling and inducing stress acclimatization. GO Biological Process acts upstream of or within regulation of gene expression, response to reactive oxygen species. GO Biological Process **involved in protein quality control for misfolded or incompletely synthesized proteins**

Protein Quality Control

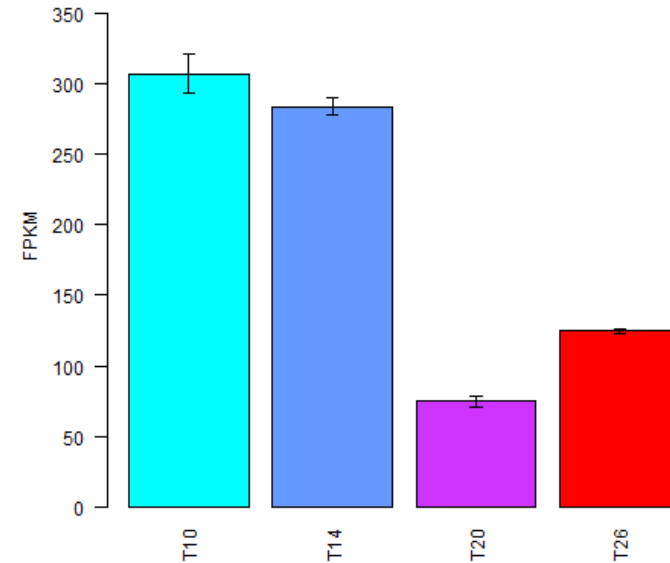


- *ostta02g01460* **Proteasome A-type subunit**
- AT1G47250.120S PROTEASOME ALPHA SUBUNIT F2, PAF2

Protein Quality Control

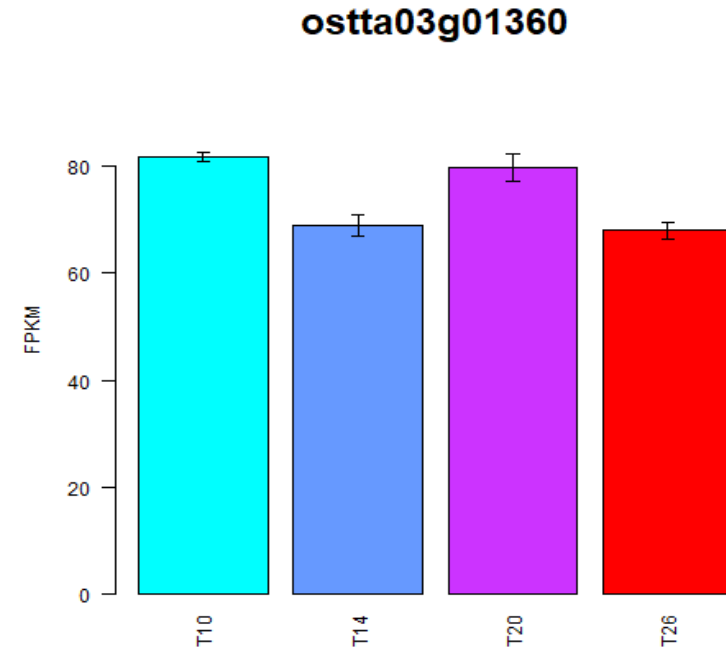
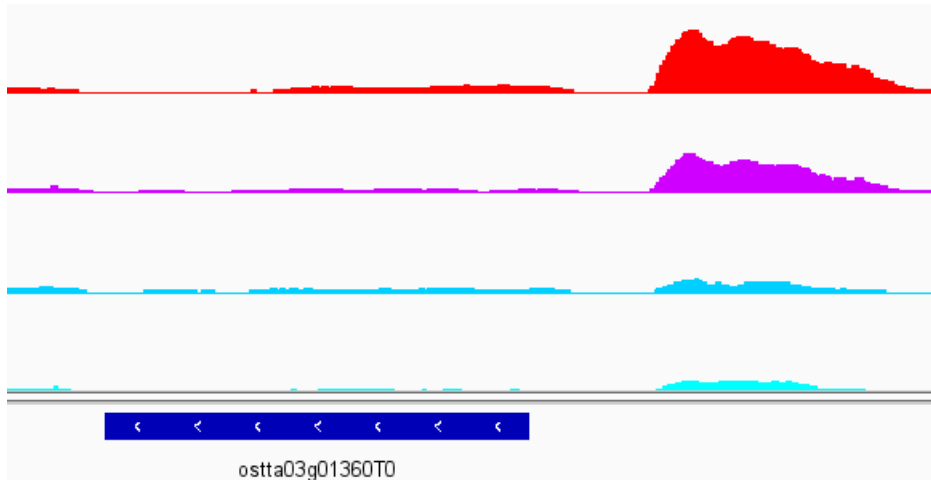


ostta10g00190

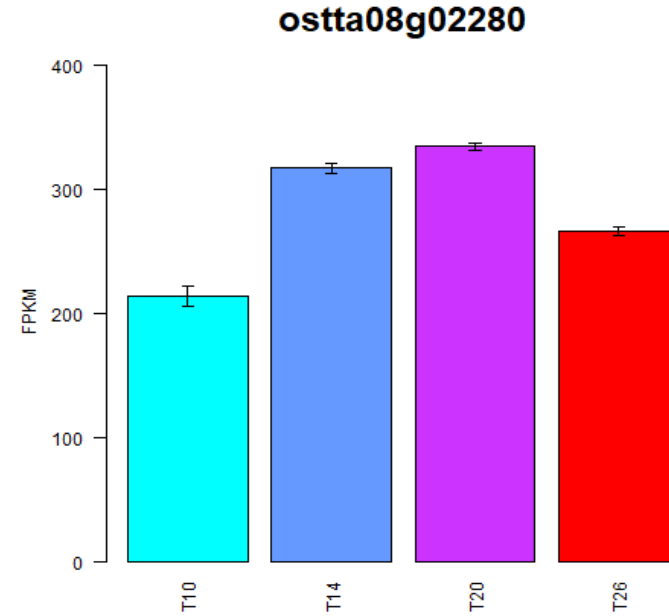
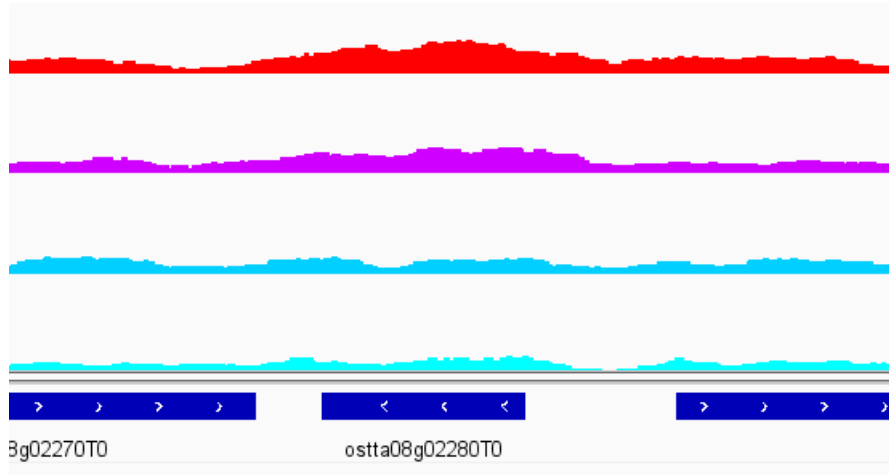


ostta10g00190 Zinc finger, RING-CH-type

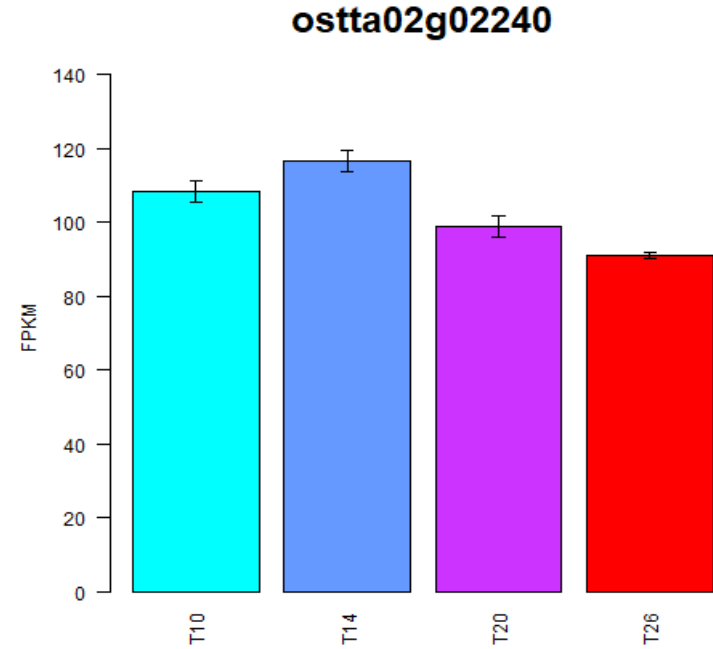
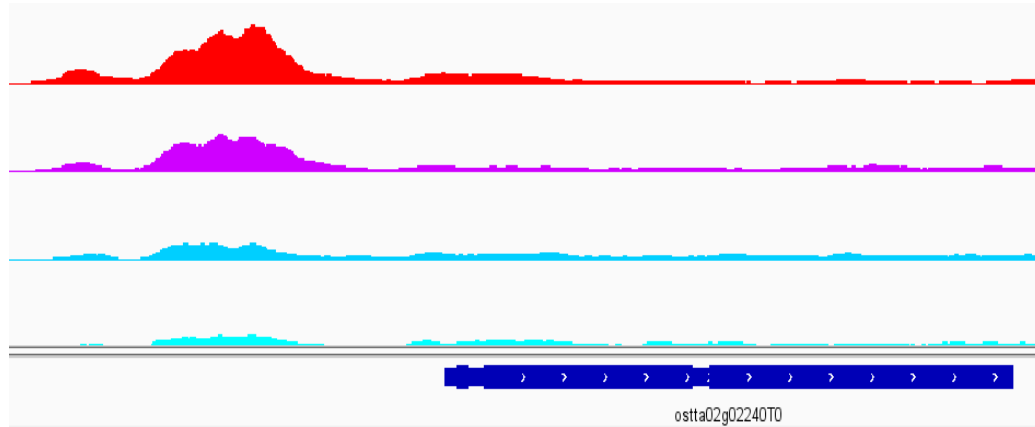
AT3G47550 GO Biological Process autophagy, cellular response to hormone stimulus, cellular response to oxygen-containing compound, defense response to bacterium, defense response to fungus, leaf senescence, protein modification process, response to lipid, response to temperature stimulus, response to water, signal transduction, transport. **Involved in Protein ubiquitination**



- ostta03g01360 unnamed product
- AT3G10800.1 **Encodes bZIP28**, a putative membrane-tethered **transcriptional factor**. **Up-regulated in response to heat**; a bZIP28 null mutant has a heat-sensitive phenotype.

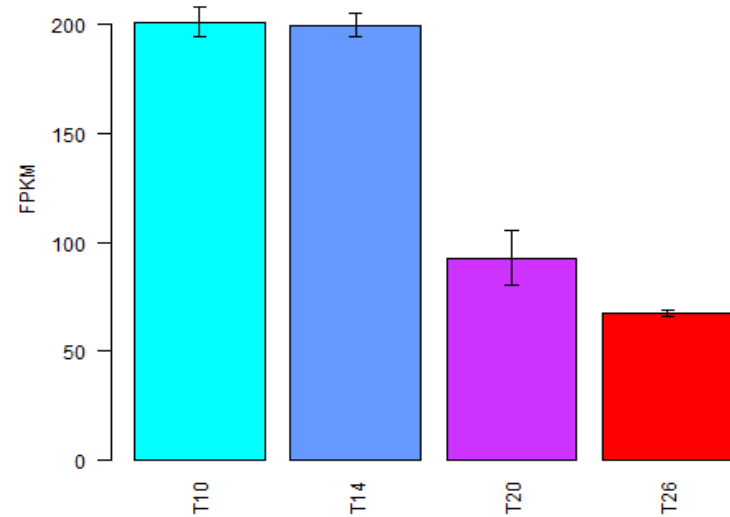
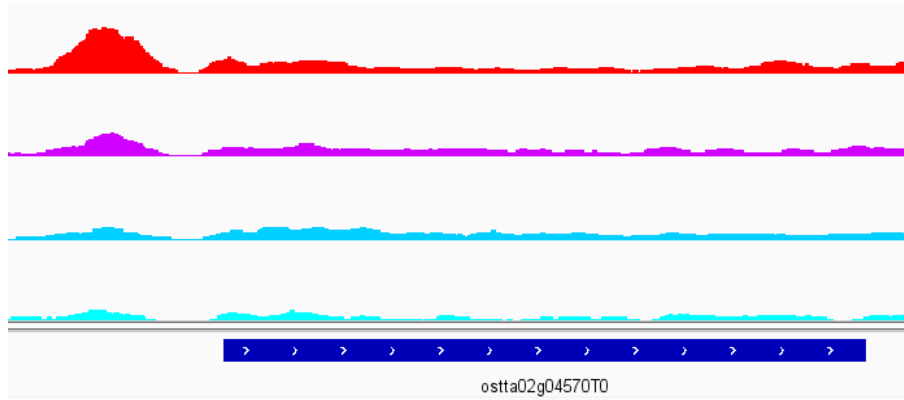


- ostta08g02280 Succinate dehydrogenase [ubiquinone] cytochrome b small subunit, CybS
- AT3G03540.1 Encodes a nonspecific phospholipase C. Located in the cytosol. Involved in the **conversion of phospholipids to glycolipids under phosphate deprivation conditions**. DNA chip-based expression profile analysis indicates involvement of the **phosphatidylinositol signaling pathway** in multiple plant **responses to hormone and abiotic treatments**.

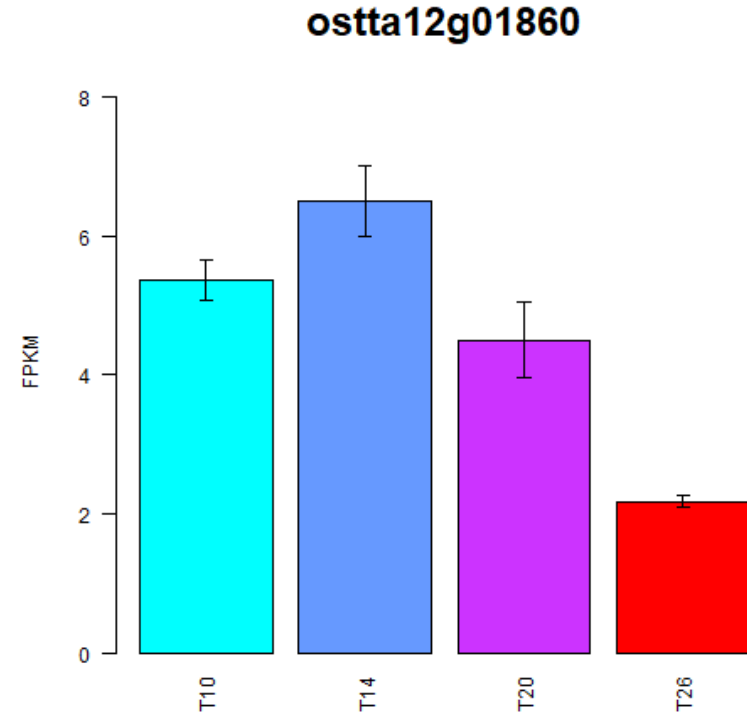
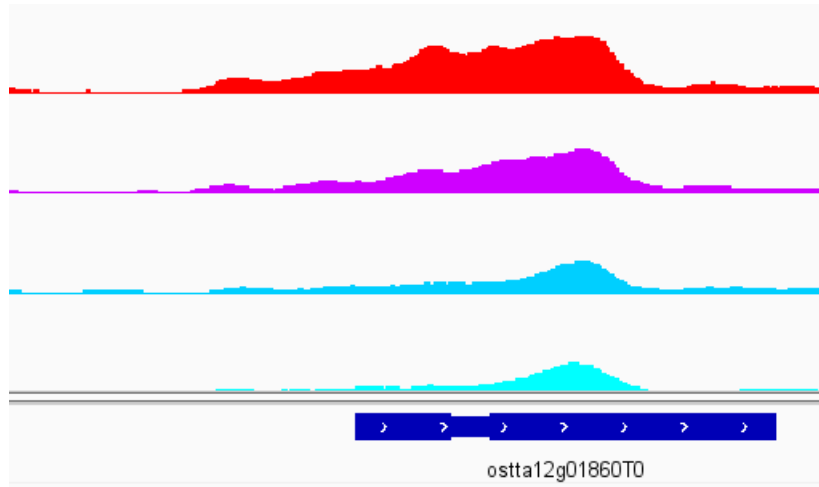


- *ostta02g02240* Pyridoxal phosphate-dependent transferase, major region, subdomain 1
- AT1G27980.1 Encodes an ER-localized **sphingoid long-chain base-1-phosphate lyase** involved in the **dehydration stress response**.

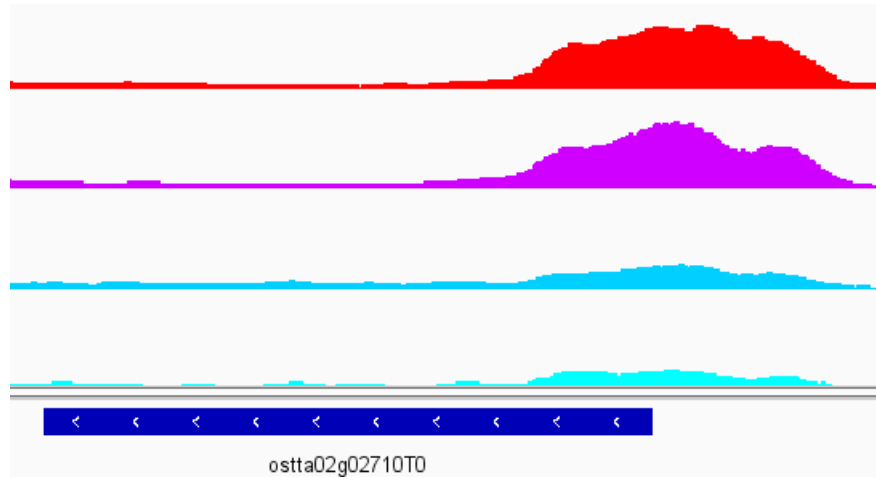
ostta02g04570



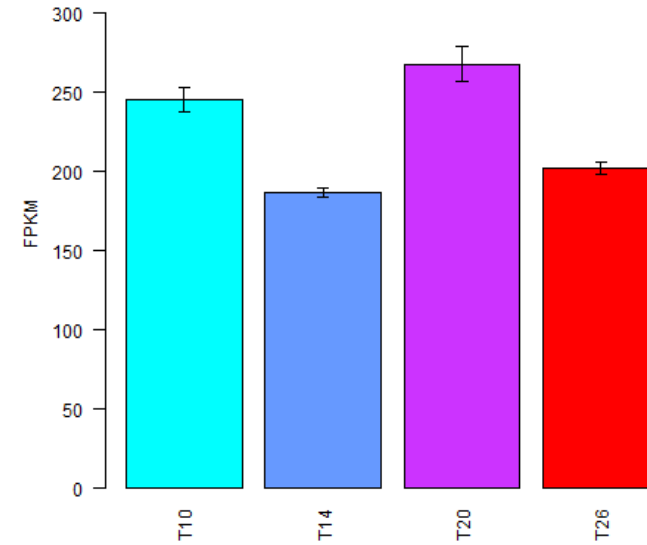
- ostta02g04570 Glycoside hydrolase, catalytic domain
- AT4G17090.1 Encodes a beta-amylase targeted to the chloroplast. Transgenic BMY8 RNAi lines fail to **accumulate maltose during cold shock** suggesting that maltose accumulation coincides with BMY8 expression.



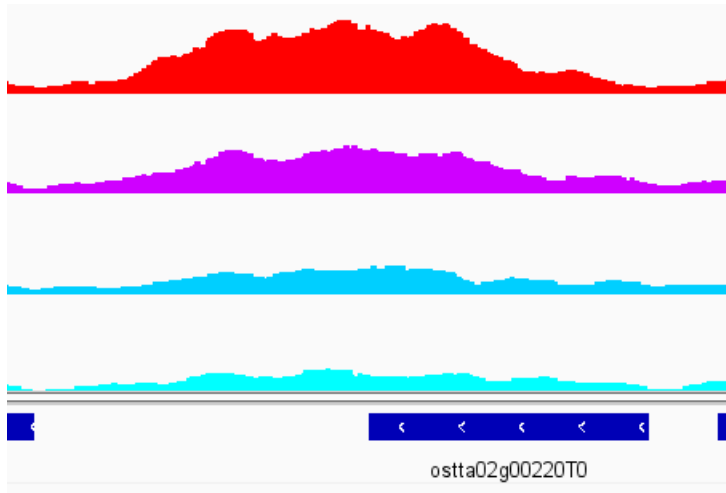
- ostta12g01860
- AT4G37930.1 Encodes a protein with mitochondrial serine hydroxymethyltransferase activity, which functions in the photorespiratory pathway, catalyzes the conversion of serine and tetrahydrofolate to glycine and 5,10-methylene tetrahydrofolate. **Involved in controlling cell damage caused by abiotic stress**, such as high light and salt and the hypersensitive defense response of plants.



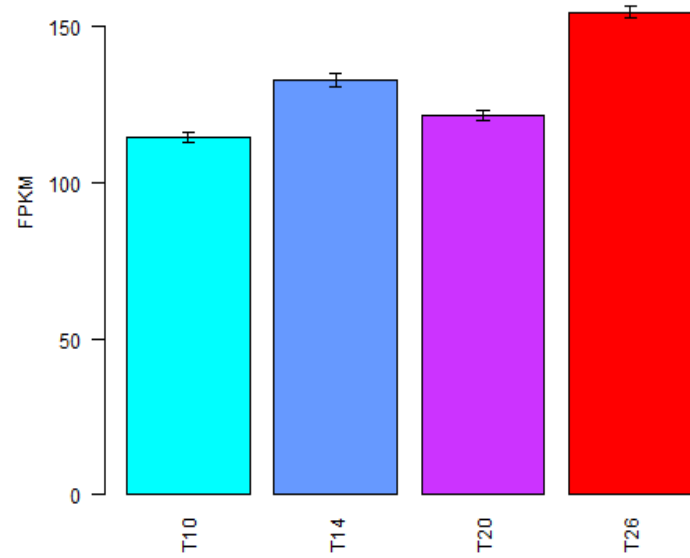
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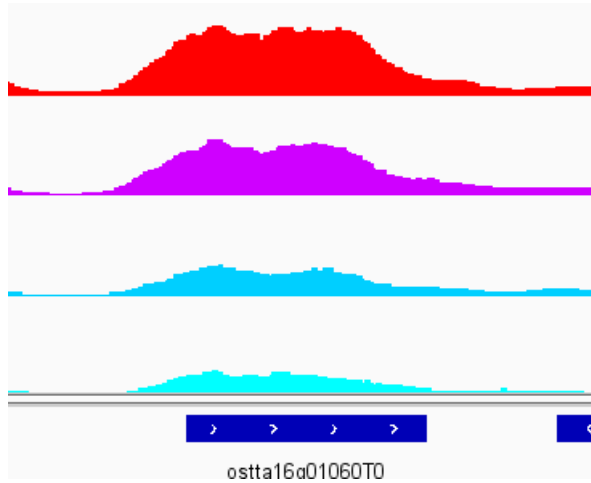
- ostta02g02710 UbiA prenyltransferase family
- AT2G18950 Encodes **homogentisate phytoltransferase involved in tocopherol biosynthesis**. Has impact on seed longevity and plays a **role in the adaptation to low temperature stress**, notably phloem loading.



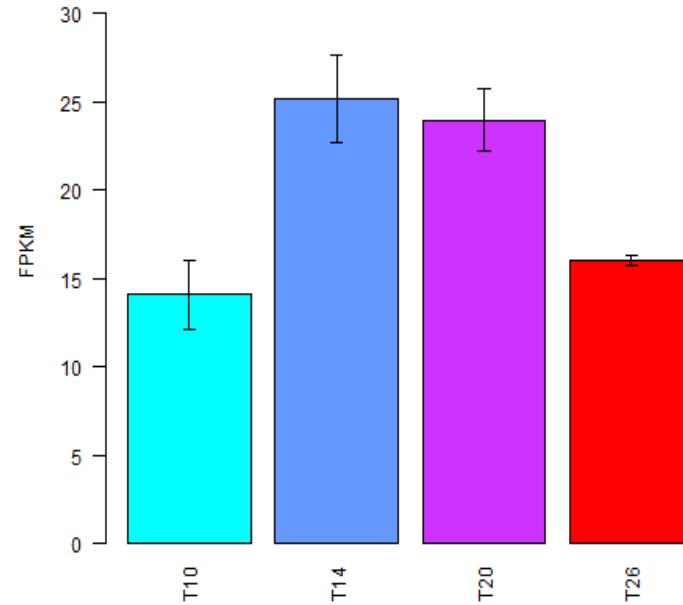
ostta02g00220



- *ostta02g00220* unnamed product



ostta16g01060



- ostta16g01060 Histone core
- AT3G20670.1 Encodes HTA13, a histone H2A protein.